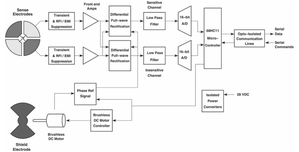
10 PRINT CHR$(205.5+RND(1)); : GOTO 10

- A one-liner BASIC program for the Commodore 64.
- Generates an infinite maze-like pattern using / and \ characters.

How it works:
- RND(1): Generates a random number between 0 and 1.
- CHR$(205.5+RND(1)): Chooses between two ASCII characters (205 = \, 206 = /).
- GOTO 10: Loops back to the first line, repeating indefinitely.

- Minimal Coding
- Early Procedural Generation

FIELDMILL

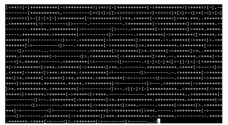


FieldMill is a technique that collects electrostatic atmospheric noise (from things like lightning strikes and ionospheric activity) to generate true random numbers.
This method has been explored for cryptography, as atmospheric electrical fluctuations are unpredictable and difficult to manipulate.

- Unpredictable Source: Atmospheric noise is inherently random and unpredictable, making it a reliable source for true randomness.
- Detection Method: By tuning a radio receiver to an unused frequency, one can capture this ambient noise. The variations in the signal can then be digitized to produce random numbers.
- Applications: True random numbers generated from atmospheric noise are used in:
- Cryptography: Creating secure encryption keys that are resistant to prediction.
- Simulations: Ensuring unpredictability in stochastic models.
- Gaming: Providing fairness in games of chance.

PIET

BRAINF#&K (.bf)

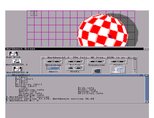


- An esoteric programming language (esolang) designed for minimalism & obfuscation.
- Created by Urban Müller in 1993.
- Known for being extremely difficult to read/write but Turing complete.

How does it work?

- Memory model: Uses an array of memory cells (like a Turing machine tape).
- Pointer-based operations: You move a pointer left/right and modify values at each cell.

Only 8 commands:



> Move pointer right
< Move pointer left
+ Increment current cell
- Decrement current cell
. Output ASCII character from current cell
, Input ASCII character into current cell
[Jump past matching] if current cell is 0 (loop start)
] Jump back to matching [if current cell is NOT 0 (loop end)

- Goal: Language with the smallest possible compiler
- 170 Bytes
- Amiga 2.0 OS
- Minimal Coding
- Language as Art

https://youtu.be/BrOJXvuzWEI?t=216

MUSE EEG

- Turns brain activity into a creative controller, No physical interaction needed.
- Blurs the line between internal & external stimuli, Music/visuals react to mental state.

Tracks four brainwave states:

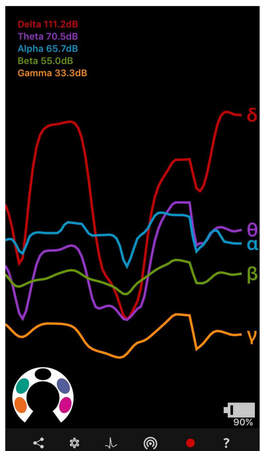
- Delta (Deep sleep, unconscious processing)
- Theta (Creativity, daydreaming, relaxation)
- Alpha (Calm focus, meditative state)
- Beta/Gamma (Alertness, problem-solving, high cognition)

Data Input:

- Muse sends EEG data via Bluetooth to a computer or phone.
- EEG values are converted into OSC (Open Sound Control) signals.

Processing in TouchDesigner:

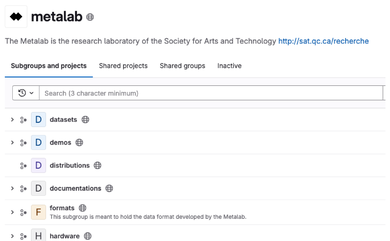
- Use an OSC In CHOP to receive brainwave data.
- Map EEG signals to visual elements, audio modulation, or FX parameters.
- Apply real-time data smoothing for stable inputs.



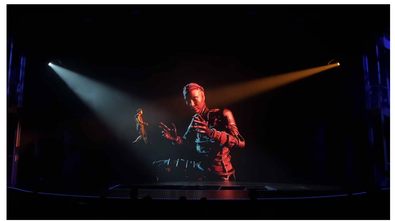
SATOSPHERE

- 360° immersive dome in Montreal, used for experimental dome projection mapping, generative visuals, and live AV performances.
- One of the first permanent domes dedicated to artistic and interactive projection experiences.
- Industry standard for dome projection setups in artistic and performance contexts.

SAT offers a TouchDesigner SDK & tools for dome projection:
https://gitlab.com/sat-mtl/metalab



BLACKTRAX



- A real-time motion tracking system for performances, projection mapping, and interactive installations.
- Uses infrared cameras and LED beacons to track performers or objects in 3D space.
- Provides precise positional data to control lighting, visuals, and stage effects dynamically

- Enhances live performances by making visuals more interactive.
- Reduces pre-programming

- Can be used with tools such as TouchDesigner and Unreal Engine

- Gaming X/R
- DJing and Musician Performances